CENTRAL MANUFACTURING TECHNOLOGY INSTITUTE

C – Smart Manufacturing Cell

User Guide

Troubleshooting - SMTP

PROJECT: SPV Data Flow

This document is confidential and intended for authorized recipients only.

1. Purpose

This document provides step-by-step troubleshooting actions when otp emails sending fails from SPV Data Flow application using SMTP. Follow each step in sequence and stop when the issue is identified.

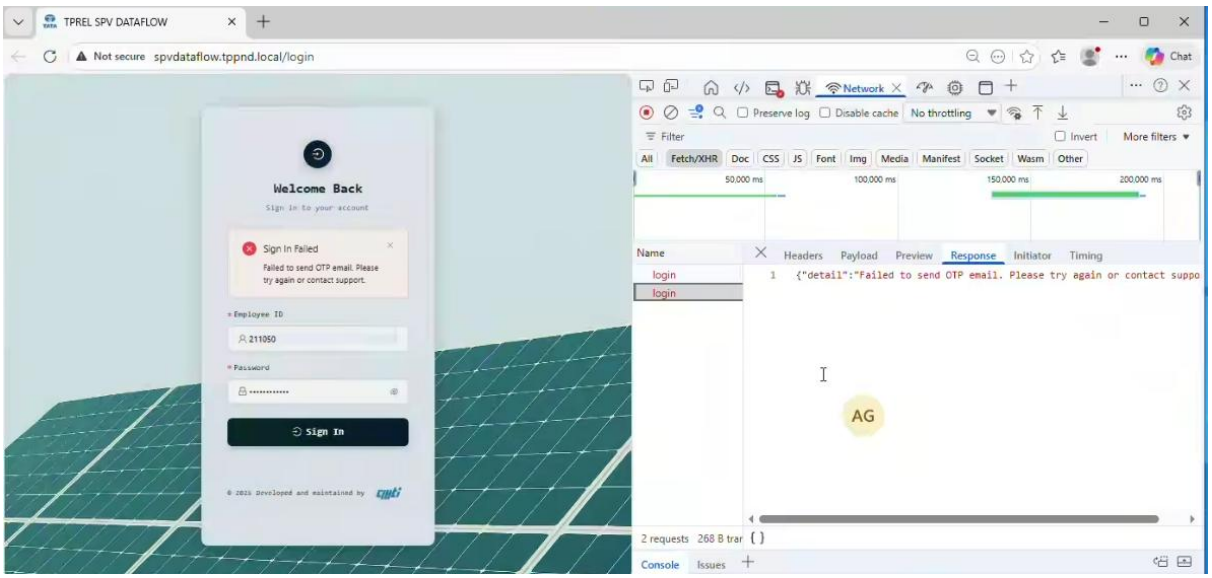
Configuration details

```
SMTP_SERVER= "172.16.200.203"
    SMTP_PORT = 25  # no TLS / no auth
    SENDER_EMAIL = "noreply@tatapower.com"
    SMTP_TIMEOUT = 15
```

2. Identify the Issue

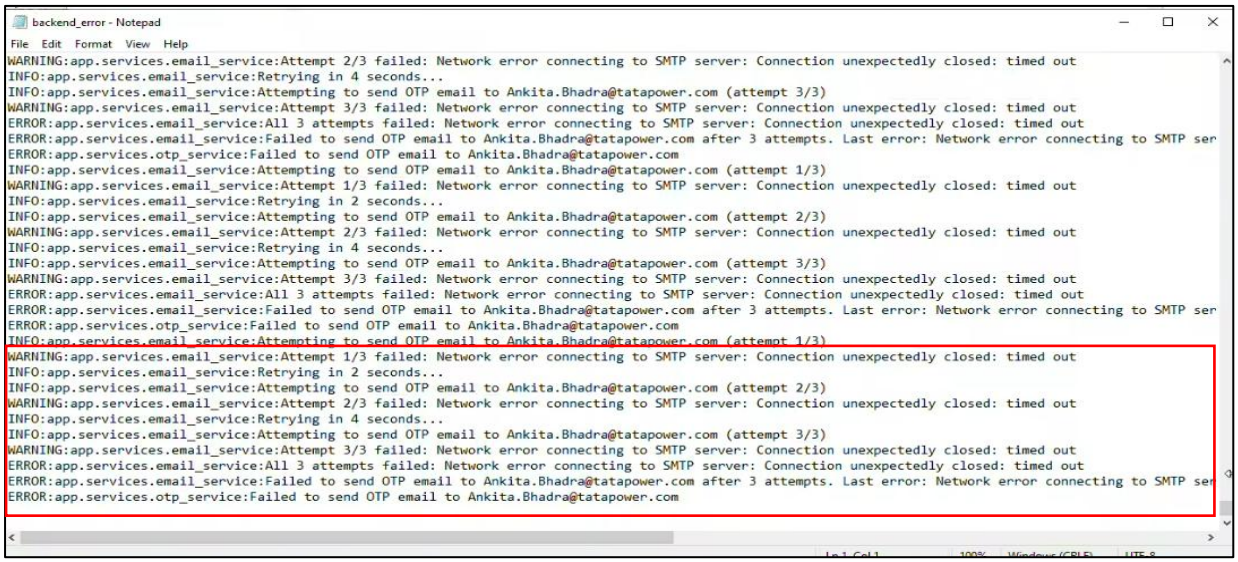
Symptoms

- Login fails because OTP emails are not being sent to user



- Application error is logged:

```
Connection unexpectedly closed: timed out
```



- Check if otp is being generated from the application

auth.otps/tatapower/postgres@PostgreSQL 17

No limit

Query | Query History

```
SELECT * FROM auth.otps
ORDER BY id ASC
```

Data Output | Messages | Notifications

	id [PK] integer	user_id integer	otp_code character varying (6)	email character varying (255)	expires_at timestamp without time zone	is_used boolean	created_at timestamp without time zone
1	1	23	811446	Ankita.Bhadra@tatapower.com	2025-12-19 07:07:13.131935	true	2025-12-19 07:02:13.132861
2	2	23	443940	Ankita.Bhadra@tatapower.com	2025-12-19 07:07:32.906826	true	2025-12-19 07:02:32.907123
3	3	23	945913	Ankita.Bhadra@tatapower.com	2025-12-19 07:13:47.988851	true	2025-12-19 07:08:47.989784
4	4	23	385780	Ankita.Bhadra@tatapower.com	2025-12-19 07:20:01.070647	true	2025-12-19 07:15:01.071587
5	5	23	693337	Ankita.Bhadra@tatapower.com	2025-12-19 07:38:41.878383	true	2025-12-19 07:33:41.879587
6	6	23	177750	Ankita.Bhadra@tatapower.com	2025-12-19 09:15:40.974934	true	2025-12-19 09:10:40.976076
7	7	23	351607	Ankita.Bhadra@tatapower.com	2025-12-22 06:36:05.756128	true	2025-12-22 06:31:05.757233
8	8	23	867074	Ankita.Bhadra@tatapower.com	2025-12-22 07:15:12.367048	true	2025-12-22 07:10:12.367384
9	9	23	105653	Ankita.Bhadra@tatapower.com	2025-12-22 10:49:38.813175	true	2025-12-22 10:44:38.813497
10	10	23	713764	Ankita.Bhadra@tatapower.com	2025-12-22 10:54:10.963155	true	2025-12-22 10:49:10.963468

3. Step-by-Step Troubleshooting

Step 1: Check Network Connectivity

Run the following commands to verify basic reachability to the SMTP server:

```
ping <SMTP_IP>
```

Administrator: Command Prompt - ping 172.16.200.203

```
C:\Windows\system32>ping 172.16.200.203

Pinging 172.16.200.203 with 32 bytes of data:
Request timed out.
```

```
tracert <SMTP_IP>
```

Administrator: Command Prompt - tracert 172.16.200.203

```
C:\Windows\system32>tracert 172.16.200.203

Tracing route to 172.16.200.203 over a maximum of 30 hops

  1     *        *
```

Expected results:

- Ping completes successfully
- Tracert reaches the destination host

Usually Ping and tracert are blocked on 10.10.12.73

If either command fails:

- This indicates a network, routing or firewall issue. Contact the respective team for further investigation.

Step 2: Check Port Connectivity

Test whether the SMTP port is reachable using Telnet:

```
telnet <SMTP_IP> 25
```

Interpret the outcome using the table below:

Case	Result	Action
A	**Connection fails**	Error: "Could not open connection" — Port is blocked or firewall is preventing access.
B	**Banner shown or Blank screen**	• If blank screen appears , TCP is working fine (no network issues , but smtp handshake not acknowleged) • "220 <hostname> ESMTP" is displayed — SMTP is working • Usually, blank screen appears on 10.10.12.73 indicating the traffic has reached firewall

Step 3: Verify SMTP Handshake

If you see a blank screen after connecting via Telnet, attempt to initiate the SMTP handshake manually. Type the following command (it may not be visible as you type):

```
EHLO test.com
```

Interpret the response:

- No response received: SMTP service is not responding, or traffic is not reaching the service. Proceed to Step 4.
- Response received (250 OK): SMTP service is working. The issue is likely in the application configuration.

Step 4: Ask Firewall Team to check firewall logs

Review firewall logs for the relevant traffic and check:

- Is the traffic from the source IP being allowed?
- Is SMTP protocol being detected or inspected?
- Are bytes flowing in both directions?

Interpret the findings:

- No traffic visible in logs: Likely a routing issue or an upstream firewall block.
- Traffic is allowed but no SMTP logs appear on the server: Traffic is not reaching the SMTP service. Investigate the following possible causes:
  - NAT or VIP misconfiguration
  - Incorrect routing rules
  - Deep packet inspection interfering with SMTP
  - Load balancer misconfiguration or failure

Step 5: Check SMTP Server Side

Engage the SMTP team and request that they check the following on the server:

- Is the SMTP service currently running?
- Is the receive connector active and enabled?
- Is the source IP address whitelisted?
- Are connection logs being generated for the source IP?

4. Escalation Summary

Use the table below to quickly determine the responsible team based on where the issue is found:

Finding	Team to Contact	Reference Step
Ping / tracert fails	Network Team	Step 1
Port blocked / connection refused	Firewall / Security Team	Step 2
SMTP handshake not completing	Firewall / Network Team	Steps 3–4
Service not running or IP not whitelisted	SMTP Team	Step 5
Connector misconfigured	SMTP Team	Step 5

If issue has been resolved , then retest by logging to the application and confirming if user received the otps to their mails,

Internal Use Only — SMTP Troubleshooting Runbook